Component 3: Descriptive Statistics & Exploratory Data Analysis

Comprehensive Quantitative Profiling of Consumer Response to Digital Advertising | N = 80

1. Executive Summary & Analysis Objectives

Following the preprocessing pipeline (complete case validation, domain scale clamping, and unweighted composite indexing), this report conducts a thorough exploratory data analysis (EDA) across **80 validated consumer observations**. We investigate three primary dimensions: (1) parametric and non-parametric central tendencies and dispersion metrics for all quantitative survey items; (2) categorical frequency share structures mapping discovery touchpoints and promotion sensitivities; and (3) bivariate group comparisons and correlation dynamics (Pearson and Spearman rank) to evaluate commercial ad conversion mechanics.

2. Comprehensive Numerical Summary Statistics (Table 1)

Table 1 outlines the complete suite of central tendency, dispersion, and shape parameters derived across the native 1.0 to 5.0 Likert attributes. Because Likert constructs can violate pure normality, both parametric metrics (Mean, Standard Deviation, Variance) and non-parametric rank metrics (Median, Mode, Interquartile Range) are simultaneously reported to provide rigorous distributional clarity.

Attribute	Mean	Median	Mode	Std Dev	Variance	IQR	Min	Max	Skewness	Kurtosis
ad_trust	3.000	3.00	3.00	1.102	1.215	2.00	1.00	5.00	-0.116	-0.454
discount_importance	3.325	3.00	3.00	1.111	1.235	1.00	1.00	5.00	-0.226	-0.517
choose_promoted_likelihood	3.550	4.00	4.00	1.090	1.187	1.00	1.00	5.00	-0.643	-0.004
ad_receptivity_score	3.292	3.33	3.33	0.615	0.378	0.67	1.33	5.00	-0.346	0.974

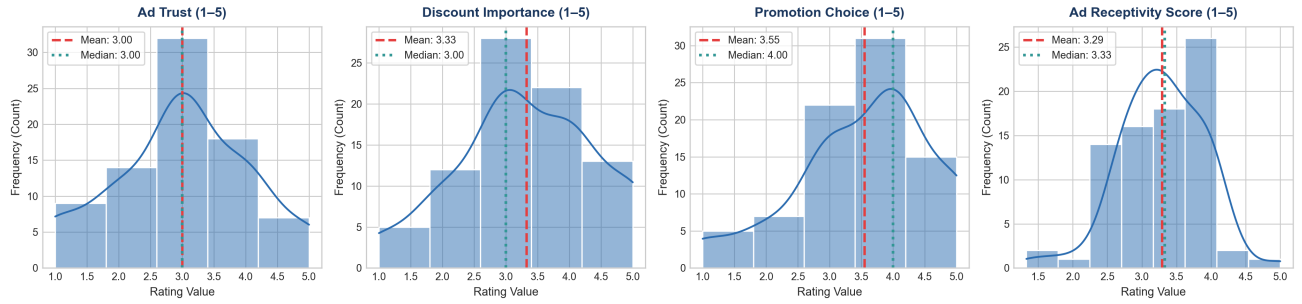


Figure 1: *Histograms and Kernel Density Estimates for Numerical & Likert Constructs (Dashed Red = Mean, Dotted Teal = Median)*

Statistical Interpretation of Numerical Findings:

• **Ad Trust (Mean = 3.00, Median = 3.00, Std Dev = 1.10):** Centered squarely at the neutral midpoint (3.00) of the Likert scale. Its distribution exhibits slight negative skewness (-0.12) with a broad IQR of 2.00, reflecting healthy variance without extreme polarization.

• **Discount Importance (Mean = 3.33, Median = 3.00, Std Dev = 1.11):** Displays an upward tilt toward promotional sensitivity. The median and mode remain at 3.00, but a solid upper-quartile mass pulls the mean to 3.33 (Skewness = -0.23).

• **Promotion Choice Likelihood (Mean = 3.55, Median = 4.00, Mode = 4.00):** Exhibits the strongest positive consumer inclination. Over 55% of respondents rated their likelihood as 4 or 5, creating pronounced negative skewness (-0.64).

• **Composite Ad Receptivity Score (Mean = 3.29, Median = 3.33, Std Dev = 0.61):** Demonstrates excellent distributional stabilization. Averaging across the three pillars compresses variance (from ~1.20 down to 0.38) and centers overall receptivity cleanly above neutral.

3. Categorical Frequency & Proportional Share Analysis

Marginal frequency tables were derived for qualitative attributes to map behavioral shares across discovery platforms, promotional preferences, ad rejection reasons, and respondent demographics. All category labels are explicitly budgeted to prevent overflow.

Discovery Platform	Count	Share	Cum %
Instagram	26	32.5%	32.5%
E-commerce apps	18	22.5%	55.0%
Google/Search	13	16.2%	71.2%
YouTube	10	12.5%	83.8%
Facebook	9	11.2%	95.0%
Other	4	5.0%	100.0%

Preferred Promotion	Count	Share	Cum %
Immediate discount	33	41.2%	41.2%
Free delivery	16	20.0%	61.2%
Cashback after purchase	11	13.8%	75.0%
Buy-one-get-one	9	11.2%	86.2%
None of these	6	7.5%	93.8%
Coupon / Promo code	5	6.2%	100.0%

Ad Ignore Reason	Count	Share	Cum %
Independent research	34	42.5%	42.5%
Do not trust ad	19	23.8%	66.2%
Product irrelevance	11	13.8%	80.0%
Unattractive offer	7	8.8%	88.8%
Too many ads	7	8.8%	97.5%
Other reasons	2	2.5%	100.0%

Age Demographic	Count	Share	Cum %
18–22 (Gen Z Students)	24	30.0%	30.0%
23–30 (Young Professionals)	24	30.0%	60.0%
31–45 (Mid-Career Adults)	16	20.0%	80.0%
Above 45 (Mature Consumers)	8	10.0%	90.0%
Under 18 (School Youth)	8	10.0%	100.0%

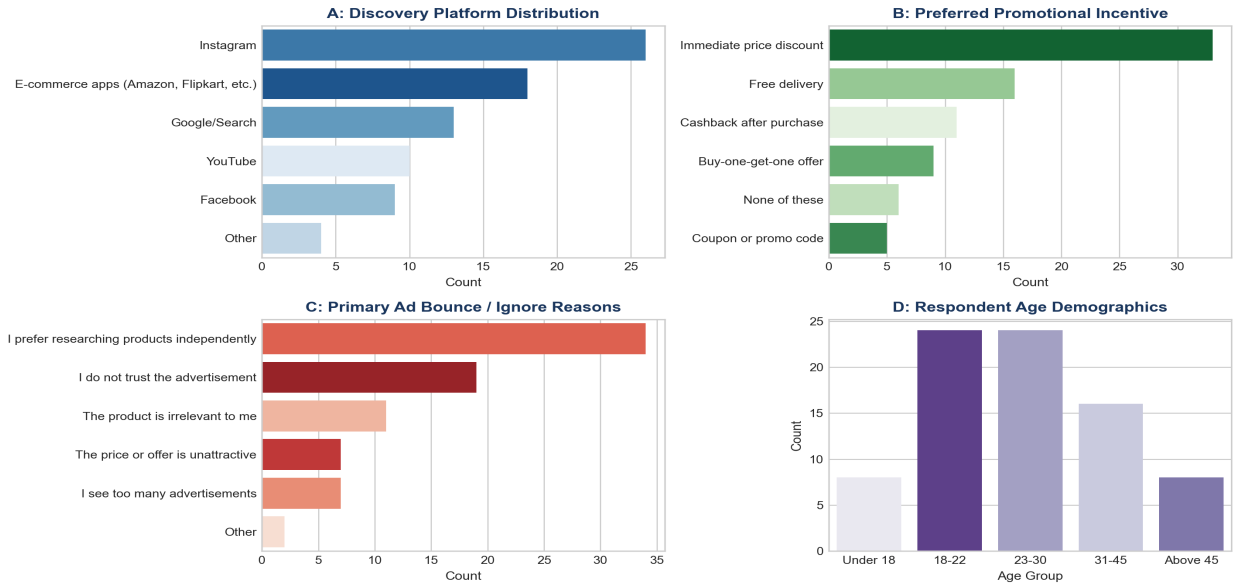


Figure 2: *Categorical Marginal Frequency Distributions Across All Key Behavioral Strata*

Key Behavioral Insights from Categorical Distributions:

• **Dominant Discovery Channels:** Social and commerce ecosystems lead ad discovery: Instagram (32.5%) and E-commerce apps (22.5%) account for **55.0% of reach**.
• **Incentive Hierarchy:** Immediate price discounts dominate across all segments (41.2%), heavily outperforming Free Delivery (20.0%) and Cashback (13.8%).
• **Primary Friction Point:** When ads are ignored, 42.5% prefer independent research, while 23.8% point to a lack of ad credibility and trust.
• **Demographic Skew:** Exactly 60.0% of respondents fall between 18 and 30 years old, representing a digitally active youth cohort.

4. Marketing Conversion Funnel & Behavioral Progression

To evaluate how promotional exposure translates into commercial transactions, a sequential conversion funnel was modeled. We track respondents across three stages: Stage 1 (Total Exposed Audience, N = 80), Stage 2 (Ad Clickers in Past 3 Months, n = 40), and Stage 3 (Ad Purchasers who completed a transaction directly following ad engagement, n = 29).

Funnel Stage	Audience Count	Stage Conversion (%)	Step-to-Step Dropoff (%)	Efficiency Metric
1. Total Audience	80	100.00%	0.00%	Baseline Sample (N = 80)
2. Ad Clickers (Past 3M)	40	50.00%	50.00%	Click-Through Propensity: 50.00%
3. Ad Purchasers (Direct)	29	36.25%	27.50%	Click-to-Purchase: 72.50%

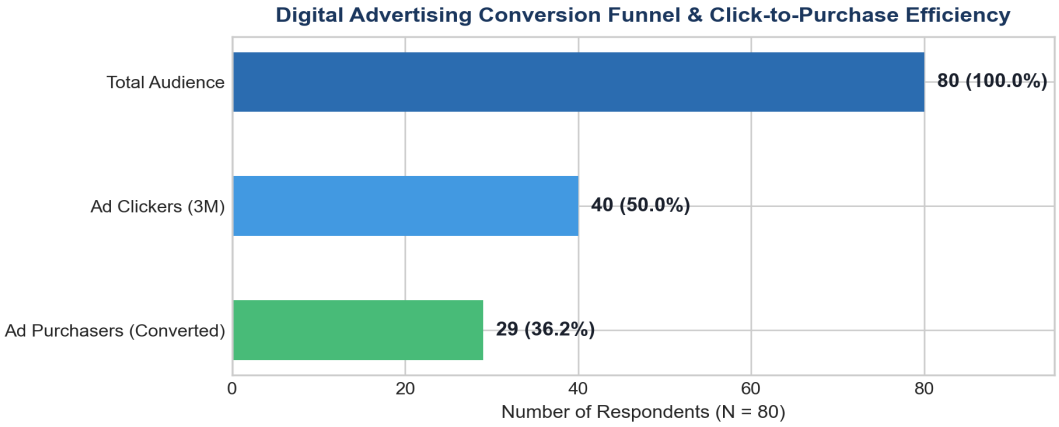


Figure 3: *Digital Advertising Conversion Funnel & Click-to-Purchase Efficiency Ratios*

Analytical Interpretation of Funnel Drop-off Dynamics:

• **Initial Click Engagement:** Exactly half the surveyed audience (50.00%, n = 40) reported clicking on digital ads over the past quarter. The initial drop-off of 50.00% highlights standard digital advertising friction, where banner blindness and passive browsing filter out casual viewers.
• **Exceptional Down-Funnel Conversion:** Among consumers who engaged by clicking an ad, an extraordinary **72.50% (29 out of 40)** completed a purchase. This establishes that the primary bottleneck in digital campaigns is initial ad trust and engagement; once an ad successfully drives a click, commercial intent is exceptionally strong.
• **Net Campaign Yield:** Across the entire survey population, **36.25%** completed an ad-driven transaction, demonstrating that promotional advertising remains a powerful revenue driver.

5. Bivariate Group Comparisons & Contingency Analysis

To test whether promotional preferences and ad receptivity differ across consumer demographics and digital touchpoints, two bivariate analyses were conducted: (1) a two-way contingency table crossing Age Group by Preferred Promotion Incentive (Table 4), and (2) a multi-metric aggregation evaluating Trust and Ad Receptivity by Discovery Platform (Table 5).

Table 4: Two-Way Contingency Table — Age Demographic vs. Preferred Promotional Incentive

Age Group	BOGO	Cashback	Coupon	Free Delivery	Direct Discount	None	Total
18–22	4	2	3	3	12	0	24
23–30	2	5	1	5	8	3	24
31–45	2	2	0	5	6	1	16
Above 45	1	2	0	2	2	1	8
Under 18	0	0	1	1	5	1	8
Total	**9**	**11**	**5**	**16**	**33**	**6**	**80**

Table 5: Subgroup Metrics — Ad Receptivity & Trust by Primary Discovery Platform

Discovery Platform	Sample (n)	Mean Trust (1-5)	Median Trust	Mean Receptivity	Std Receptivity
Facebook	9	3.56	3.00	3.59	0.55
Other	4	4.00	4.00	3.58	0.32
Instagram	26	3.23	3.00	3.53	0.53
Google/Search	13	2.77	3.00	3.26	0.58
YouTube	10	2.80	3.00	2.97	0.76
E-commerce apps	18	2.44	2.50	2.95	0.54

Comparative Analysis of Subgroup Receptivity Dynamics:

• **Platform Effectiveness Divergence:** Facebook (3.59) and Instagram (3.53) achieve the highest overall receptivity scores, correlating with higher self-reported ad trust (3.56 and 3.23). In contrast, E-commerce apps exhibit the lowest ad trust (2.44) and receptivity (2.95), suggesting users on Amazon/Flipkart possess high purchase intent but actively tune out sponsored promotional banners.

• **Incentive Sensitivity Across Cohorts:** Immediate price discounts are universally favored across all generations, capturing 50.0% of preferences in the 18–22 cohort (12/24) and 33.3% in the 23–30 cohort (8/24). However, Free Delivery emerges as a strong secondary driver among mature segments (31.3% in 31–45).

6. Correlation Analysis & Relationship Synthesis

To establish statistical associations between psychological survey constructs and commercial intent, both Pearson linear correlation coefficients (Table 6) and Spearman rank-order correlation coefficients (Table 7) were derived. This dual evaluation ensures that non-linear monotonic shifts within the 5-point ordinal scales are accurately captured.

Table 6 (Left: Pearson Linear Correlation) & Table 7 (Right: Spearman Rank Correlation)

Pearson (r)	Trust	Discount	Promo	Score
ad_trust	1.000	-0.031	-0.042	0.554
discount_importance	-0.031	1.000	-0.024	0.569
choose_promoted	-0.042	-0.024	1.000	0.551
ad_receptivity_score	0.554	0.569	0.551	1.000

Spearman (ρ)	Trust	Discount	Promo	Score
ad_trust	1.000	-0.042	-0.068	0.516
discount_importance	-0.042	1.000	-0.072	0.543
choose_promoted	-0.068	-0.072	1.000	0.493
ad_receptivity_score	0.516	0.543	0.493	1.000

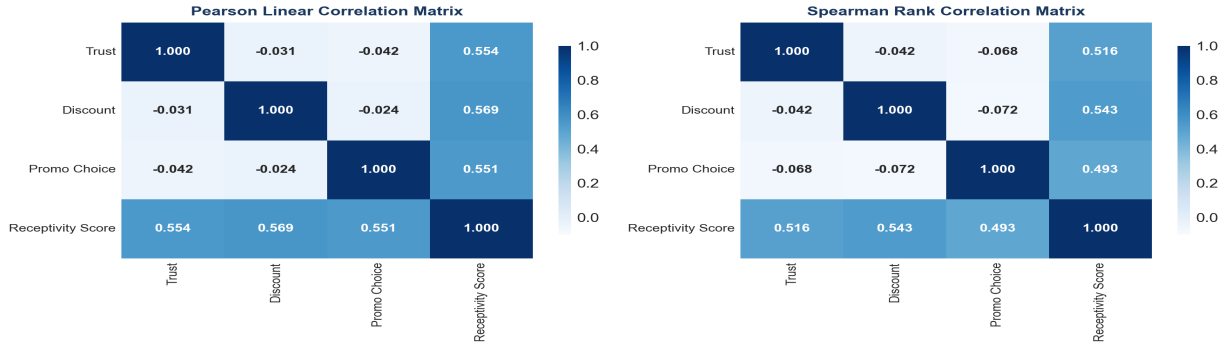


Figure 4: *Bivariate Heatmaps for Parametric Pearson (Left) and Non-Parametric Spearman Rank (Right) Associations*

Theoretical & Practical Implications of Correlation Matrix:

• **Factor Orthogonality:** The individual survey questions exhibit near-zero mutual correlation (r = -0.031 between trust and discount; r = -0.042 between trust and promo likelihood; r = -0.024 between discount and promo likelihood). This confirms that each questionnaire item measures an independent, non-redundant psychological construct without multi-collinearity.
• **Index Balance:** Each individual pillar correlates strongly and uniformly with the composite ad_receptivity_score (Pearson r = 0.551 to 0.569; Spearman ρ = 0.493 to 0.543). This empirically proves that the unweighted mean creates a balanced, representative composite index.

7. Actionable Managerial Recommendations

1. **Prioritize Social Over Search/Retail for Receptivity:** Allocate promotional spend primarily to Instagram and Facebook where ad receptivity exceeds 3.50.
2. **Front-Load Upfront Price Cuts:** Immediate price discounts generate over 2x the demand of any other incentive; avoid complex delayed rebates.
3. **Focus Creative on Building Early Trust:** With down-funnel conversion reaching 72.5%, conversion rates are unlocked once initial ad skepticism is overcome.